Introduction to Generative AI



Agenda

- Introduction to Generative AI
- Deep Dive into LLMs (Large Language Models)
- Popular Tools & Real-world Use Cases
- Prompt Engineering Basics
- Introduction to Langchain
- Ethics & Responsible AI Use
- Mini Project: Build with Gen AI

Introduction to Generative AI

" Generative AI refers to systems that can **create new content** — like text, images, music, or code — that's similar to human-created content."

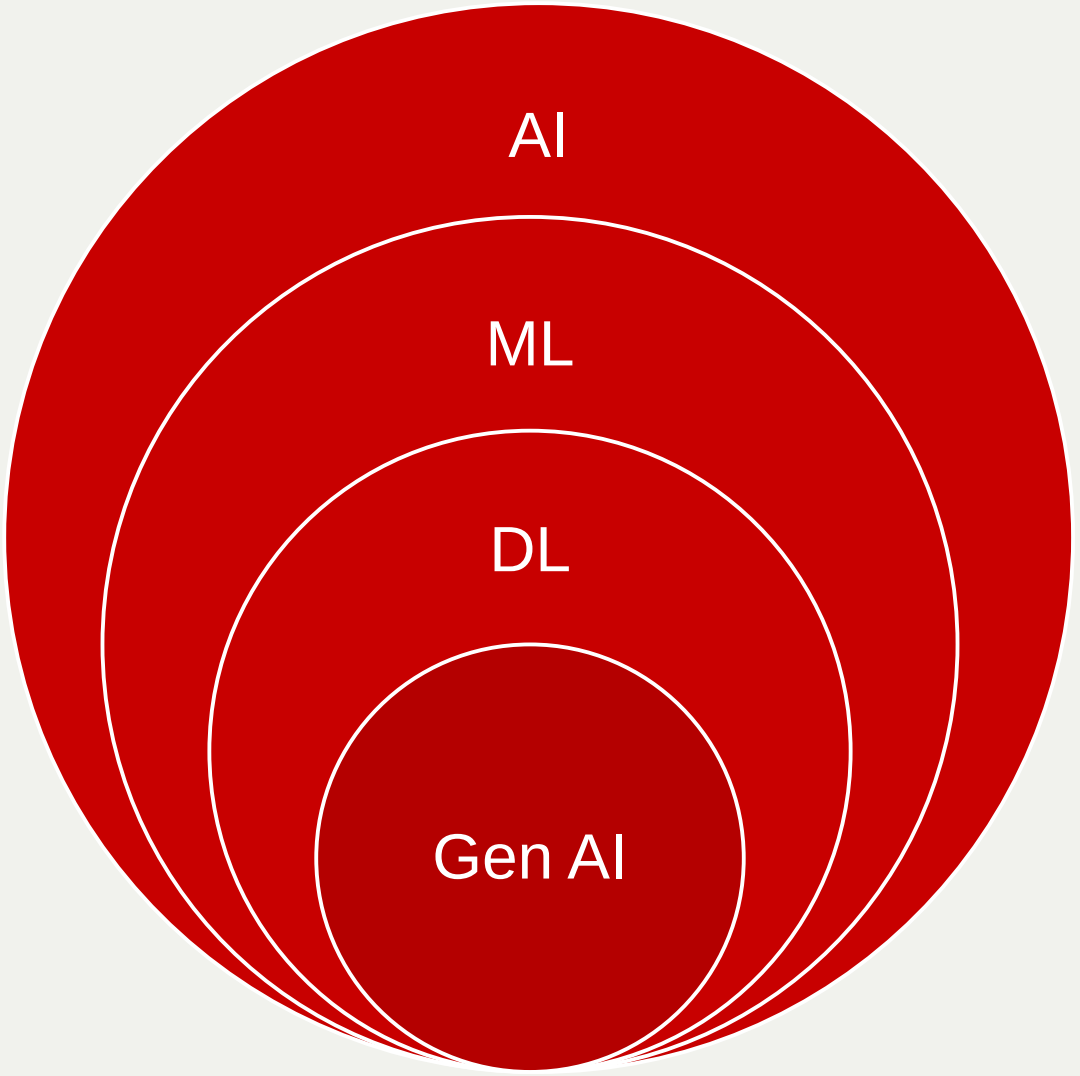
Types of Content Generated:

- Text → Blogs, emails, poems (ChatGPT, Claude)
- Images → Art, designs, product mockups (DALL·E, MidJourney)
- Audio → Music, voiceovers (Murf, MusicGen)
- Code → Functions, scripts (GitHub Copilot)
- Example- Following Images is generated by DALL-E and video is generated by Sora





Where Generative AI stands?



Where Gen AI is used?

Content Creation:

- Automated text generation for articles, blogs, and social media posts.
- Creative writing including stories, poems, and scripts.
- Image and video generation for marketing, entertainment, and art.

Customer Support:

- Chatbots and virtual assistants providing 24/7 customer service.
- Automated response generation for emails and queries.

Healthcare:

- Generation of synthetic medical data for research.
- Personalized health advice and diagnostics.

Where is Generative AI used?

Education:

- Personalized learning content and tutoring.
- Automated grading and feedback for assignments.

Entertainment and Media:

- Creation of virtual characters and environments in gaming.
- Scriptwriting and storyline development for movies and TV shows.

Finance:

- Automated financial reports and analysis.
- Fraud detection through anomaly generation and analysis.

Where is Generative AI used?

Design and Fashion:

- Creation of new fashion designs and patterns.
- Architectural design and interior decoration suggestions.

Marketing and Advertising:

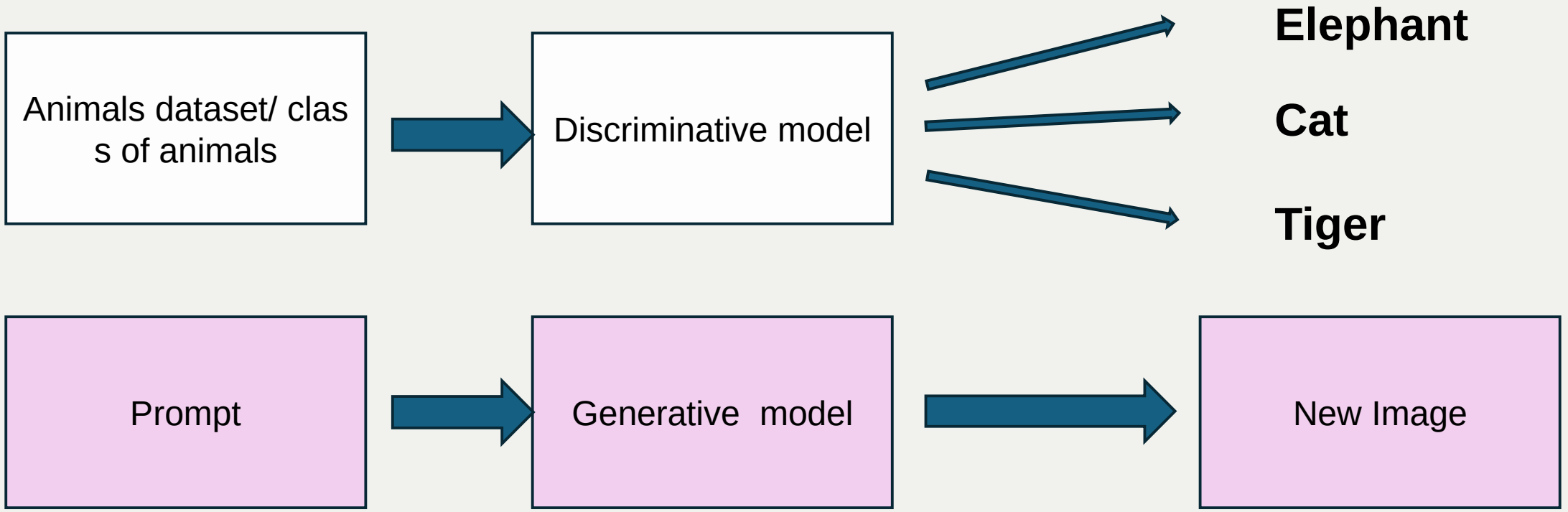
- Personalized ad content generation.
- Market analysis and trend prediction through data generation.

Manufacturing:

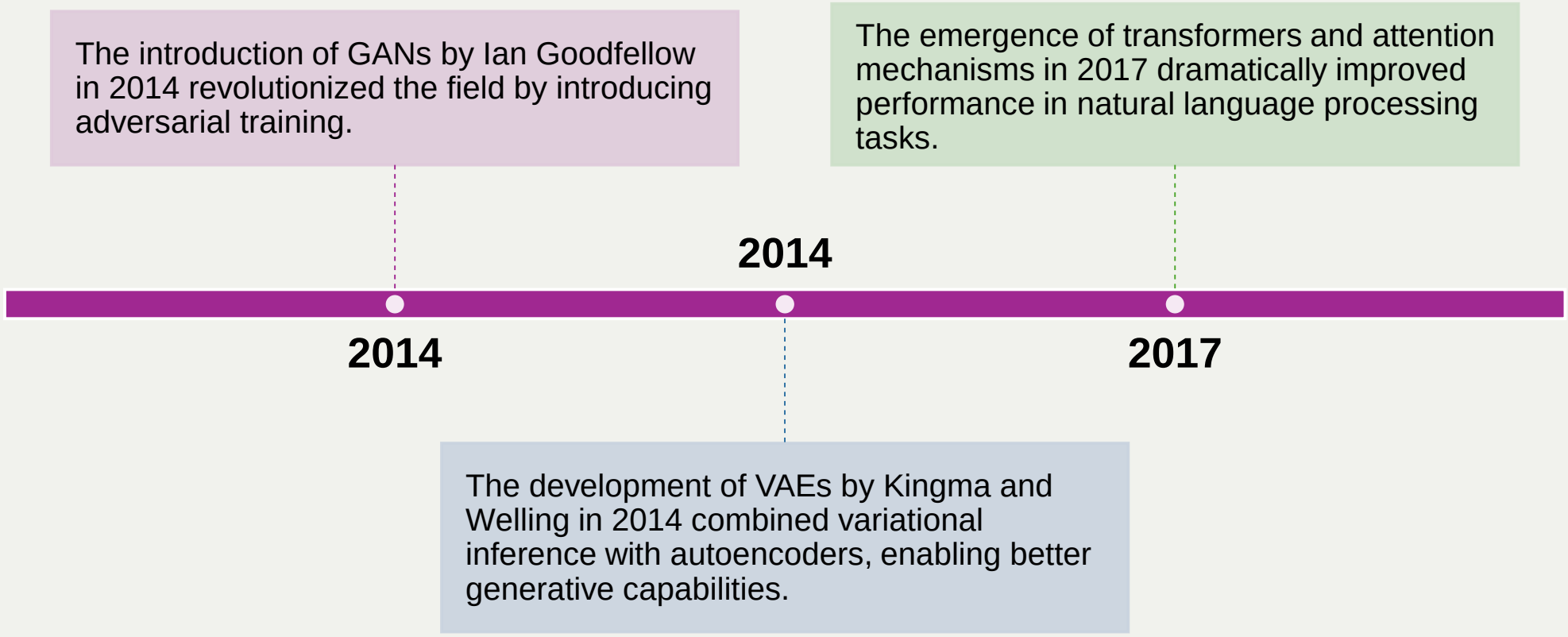
- Design and optimization of new products.
- Simulation of manufacturing processes.

Difference b/w Discriminative Vs Generative models

- **Generative** models can generate new data instances.
- **Discriminative** models discriminate between different kinds of data instances.



Key milestones in Generative AI Development



Types of Generative AI Models

Model Type	Description	Examples
Transformer-based Models	Use self-attention to understand and generate sequences. Backbone of Large Language Models (LLMs). Excellent for text, code, and multimodal tasks.	GPT (OpenAI), BERT (Google), Claude (Anthropic), LLaMA (Meta), Gemini (Google)
Generative Adversarial Networks (GANs)	Two-part system: Generator creates data, Discriminator evaluates it. Great for generating realistic images.	StyleGAN, BigGAN
Variational Autoencoders (VAEs)	Compress data into a latent space and reconstruct it. Good for smooth variations and interpolation.	VAE, Beta-VAE
Diffusion Models	Gradually remove noise from random inputs to generate high-quality outputs. Popular in image synthesis.	Stable Diffusion, DALL·E 3
Autoregressive Models	Predict next element based on previous ones. Transformers like GPT are also autoregressive in nature.	GPT-4, PixelCNN
Flow-based Models	Use reversible transformations to model data distributions directly. Less common but mathematically elegant.	RealNVP, Glow

Large Language Models (LLMs)

Large Language Models (LLMs) are advanced artificial intelligence models designed to understand, interpret, and generate human-like text based on the vast amounts of data they have been trained on.

Purpose:

- Used in chatbots, translation, content creation, and more.
- Excels in understanding context and generating human-like responses.
- Performs tasks involving complex language processing.

Core Components of LLMs

- Tokens: Basic units of text processing
- Embeddings: Representing words in vector space
- Attention Mechanism: Focusing on important parts of the input

Attention that all you need

- Attention is All You Need" stands as one of the most influential research papers in AI of the last decade.
- It catalysed the advancements in NLP, machine translation, and paving the way for the emergence of generative AI models
- The paper introduced the Transformer model, which marked a paradigm shift in natural language processing (NLP) and machine translation tasks.



Attention Is All You Need

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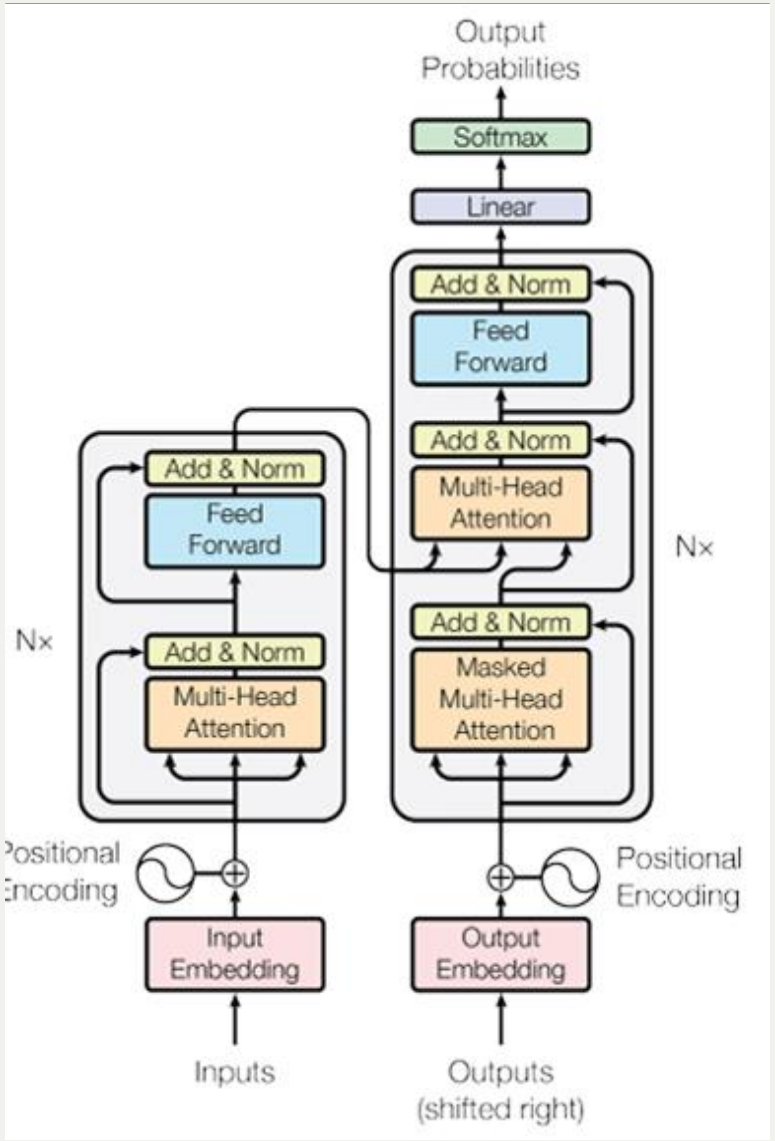
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Transformer Architecture

- Specialised architecture for token prediction
- Key information – Attention mechanism
- Not just a big neural net



From where do they get such big data???

- **Massive Training Data**
  - Utilizes an extensive dataset sourced from the entire World Wide Web
- **Scale and Investment**
  - Significant investments from major corporations, including Microsoft
  - Only large companies can afford to fund such expansive projects
- **Data Volume**
  - The training data is enormous, reflecting the vast amount of text available online
- **Role of OpenAI**
  - Pioneering advancements in AI and natural language processing through extensive research and resources

Popular Tools & Real-World Use Cases

Popular Generative AI Tools:

Text Generation:

- ChatGPT (OpenAI)
- Claude (Anthropic)
- Gemini (Google)
- Cohere, Mistral

Image Generation:

- DALL·E (OpenAI)
- MidJourney
- Leonardo AI
- Bing Image Creator

Code Generation:

- GitHub Copilot
- Replit Ghostwriter
- TabNine

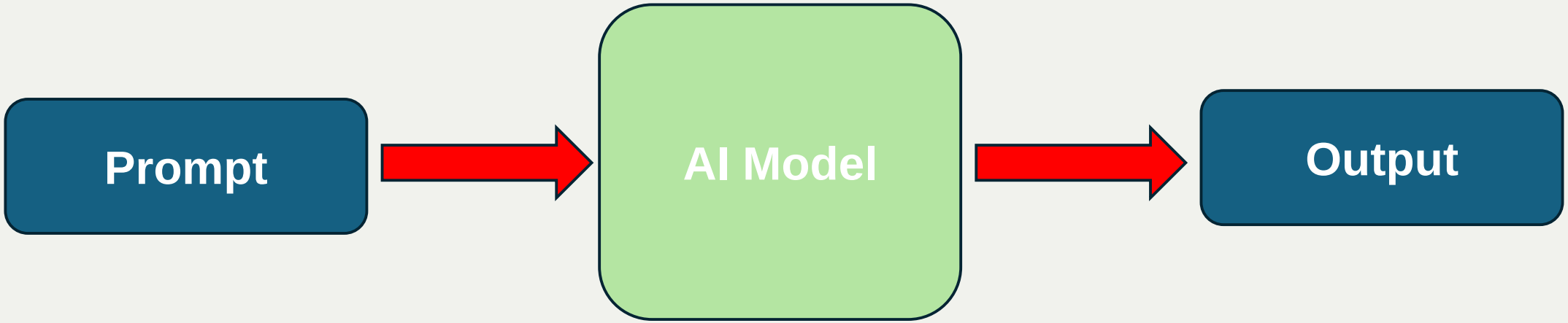
Video /Audio Generation:

- Sora
- Synthesia (video avatars)
- ElevenLabs (voice cloning)
- MusicGen / Suno AI (music)

Prompt Engineering: Getting the Best from AI

What is Prompt Engineering?

"The art and science of crafting inputs to guide AI responses."



Why Prompt Engineering Matters?

- The **quality of the prompt** directly affects the **quality of the output**
- Small changes = Big differences
- Helps you **control tone, length, format**, and more

Poorly Designed Prompt

Prompt: "Tell me about Python."

AI Response:
"Python is a type of snake. It's also a programming language."

Well-Designed Prompt

Prompt: "Explain the key features of the Python programming language used for data analysis."

AI Response:

"Python is a high-level, interpreted programming language known for its readability and versatility.

Key features include:

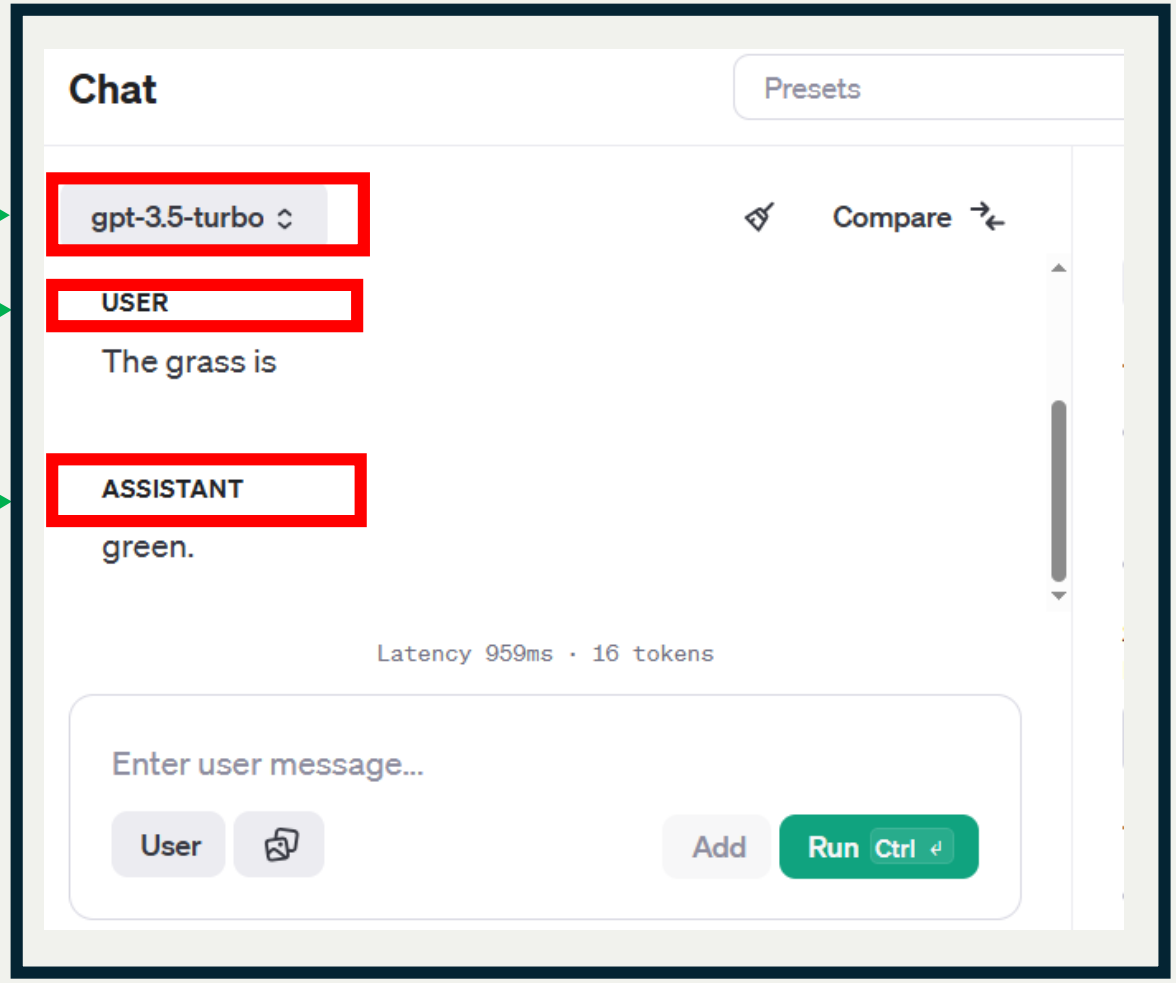
- Extensive libraries such as NumPy and pandas for data manipulation.
- Support for data visualization tools like Matplotlib and Seaborn.
- An active community and extensive documentation."

OpenAI Platform

OpenAI chat models

Prompt (Input)

Response by model



Key Elements of an Effective Prompt

⬍ Prompt Element	📝 Description	✅ Example
1. Role / Persona	Assigns an identity or expertise to the AI	"You are a senior data scientist..."
2. Task / Instruction	Clearly states what the model should do	"Summarize the article in 5 bullet points."
3. Context / Background	Provides relevant information or setup	"The article is about climate change impacts in India."
4. Input Data	The content or query to work on	"Here is the article text: [...]"
5. Format / Output Style	Specifies how the response should look	"Respond in JSON format" / "Make it a tweet-length answer"
6. Constraints	Defines limits, rules, or conditions	"Limit to 100 words" / "No technical jargon"
7. Examples (Few-shot)	Shows examples to guide model behavior	"Q: What is AI? A: AI is the science of making machines intelligent..."
8. Tone / Style	Instructs the tone of the output	"Use a friendly and engaging tone."
9. Temperature & Top-p (Advanced)	Adjusts creativity and randomness (model parameters)	temperature=0.7 for creativity; top_p=0.9

Poor Prompt (Missing Structure & Clarity)

Prompt:

Explain overfitting.

Problems with this prompt:

- ❌ No role – the model doesn't know who it's pretending to be.
- ❌ No clear task – too vague.
- ❌ No context – doesn't know the audience or purpose.
- ❌ No format – might return a long paragraph.
- ❌ No constraints – could use complex terms or be too verbose.
- ❌ No tone – could be too technical or dull.
- ❌ No example – leaves interpretation wide open.

Good Prompt (All Key Elements Used)

Prompt:

You are a senior data scientist 🧑‍🔬 **(Role)**

Your task is to explain the concept of overfitting in machine learning to a non-technical audience. □ **(Task)**

The audience consists of business stakeholders with limited technical background. 📄 **(Context)**

Explain the term using this scenario: A model that performs well on training data but poorly on new data. 📥 **(Input)**

Respond in 3 bullet points, using simple language and a relatable analogy. 📊 **(Format)**

Avoid technical jargon and keep each point under 20 words. ⊘ **(Constraints)**

Example: "Think of a student who memorized practice questions but can't answer new ones in the exam." ✨ **(Few-shot Example)**

Use a friendly and conversational tone. 😊 **(Tone)**

(Optional API setting: temperature = 0.6) 🔧 **(Advanced)**

Advanced Prompt Techniques

- Zero-shot Prompting
- Few-shot Prompting
- Chain-of-Thought (CoT) Prompting
- Role-based Prompting
- Prompt Chaining

Zero-shot prompting

Zero-shot prompting is when you ask a language model to perform a task **without providing any examples**. You rely only on a **clear and well-defined instruction**.

When to Use It:

- For **simple or well-known tasks**
- When **task is common** and the model is already trained on similar patterns
- When you want **fast results without few-shot setup**

Example

- ***Prompt:***
  “Translate the following English sentence to Hindi: ‘Artificial Intelligence is transforming the world.’”
- ***Response:***
  “कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।”

Few-Shot Prompting

Few-shot prompting is a technique where you **provide a few examples** of input-output pairs in the prompt to **guide the model** in generating similar responses.

When to Use It:

- When the task is **specific or slightly complex**
- To **demonstrate the expected format or tone**
- When zero-shot isn’t giving accurate results

Example

Prompt:

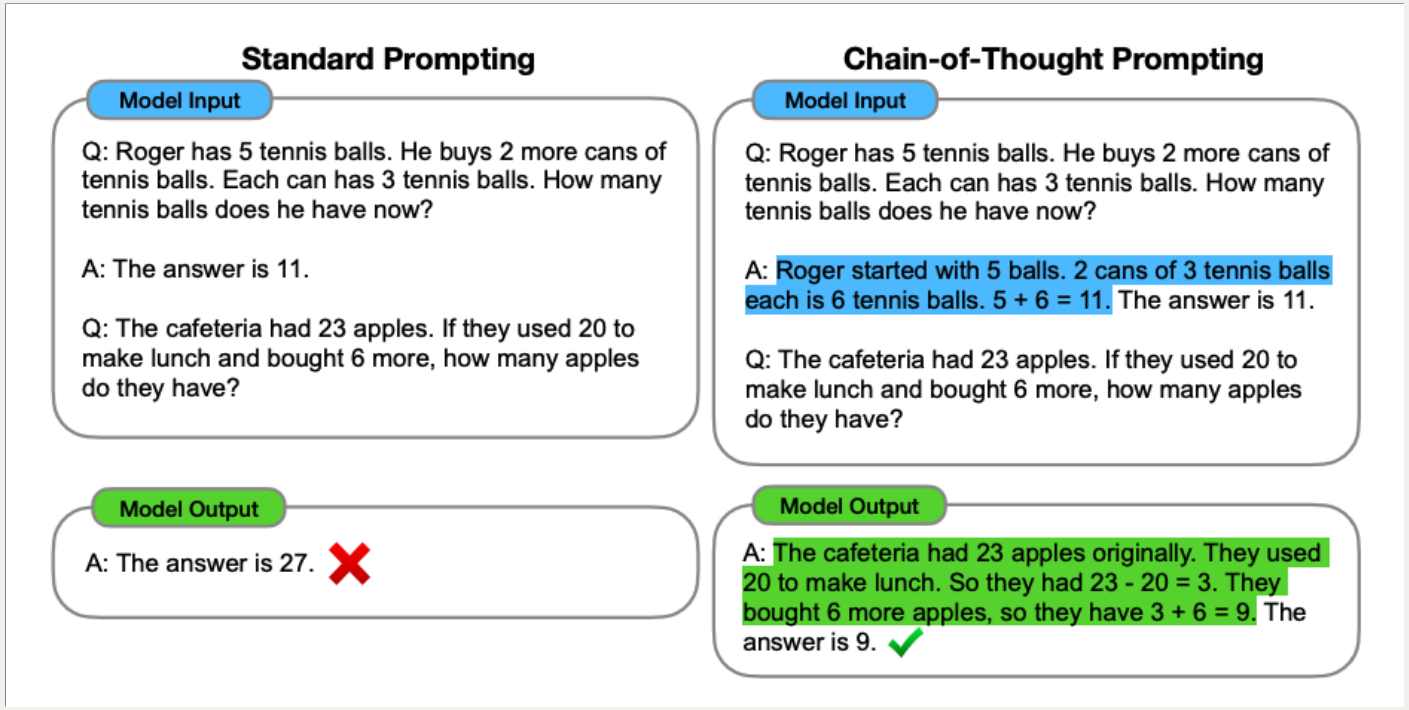
- *Q: What is AI?*
  A: It's the ability of machines to mimic human intelligence.
- Q: What is Machine Learning?
  A: It's the method that allows computers to learn from data.

Response:

A: It's a subset of ML that uses neural networks to learn from large data.

Chain-of-Thought (CoT)

- Chain-of-Thought (CoT) prompting is a technique where you guide the LLM through a step-by-step reasoning process to solve a problem.
- By breaking down the problem into smaller steps, the LLM can handle complex tasks more effectively



Chain of Thought

Prompt
The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.

The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.
A: Adding all the odd numbers (17, 19) gives 36. The answer is True.

The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24.
A: Adding all the odd numbers (11, 13) gives 24. The answer is True.

The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2.A: Adding all the odd numbers (17, 9, 13) gives 39. The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1. A:

Output: Adding all the odd numbers (15, 5, 13, 7, 1) gives 41. The answer is False.

What is Prompt Chaining?

- Prompt chaining is a technique where multiple prompts are executed in sequence, and the **output of one prompt becomes the input** for the next.
- It helps handle **complex or multi-step tasks** by breaking them down into manageable steps.

Why Use It?

- Manages **multi-stage workflows** (e.g., summarize → analyze → reformat)
- Improves accuracy and control over each task
- Enables **modular and reusable** prompt design
- Powers **Agentic AI** and **LangChain-like workflows**

Prompt chaining

Example: Writing an Essay
Task: Write an essay on the benefits of renewable energy.

- **Step 1: Generate an Outline**

Prompt: "Create an outline for an essay on the benefits of renewable energy."
Output:

- Introduction
- Definition of Renewable Energy
- Types of Renewable Energy
- Benefits of Renewable Energy
- Conclusion

Prompt chaining

- **Step 2: Write the Introduction**

Prompt: "Using the outline, write the introduction for an essay on the benefits of renewable energy."

Output: "Renewable energy is becoming increasingly important in the fight against climate change. This essay will explore the definition, types, and numerous benefits of renewable energy sources."

Prompt chaining

- **Step 3: Define Renewable Energy**

Prompt: "Write a paragraph defining renewable energy for the essay."

Output: "Renewable energy comes from natural sources that are constantly replenished, such as sunlight, wind, and water. Unlike fossil fuels, renewable energy sources do not deplete over time."

Prompt chaining

- **Step 4: Describe Types of Renewable Energy**

Prompt: "Write a paragraph about the different types of renewable energy."

Output: "The main types of renewable energy include solar, wind, hydroelectric, geothermal, and biomass energy. Each type harnesses natural processes to generate electricity or heat without emitting greenhouse gases."

Prompt chaining

- **Step 5: Explain Benefits of Renewable Energy**

Prompt: "Write a paragraph about the benefits of renewable energy."

Output: "Renewable energy offers numerous benefits, including reducing greenhouse gas emissions, decreasing air pollution, and providing sustainable energy sources. It also creates jobs and can help stabilize energy prices."

Prompt chaining

- **Step 6: Write the Conclusion**

Prompt: "Write a conclusion for the essay on the benefits of renewable energy."

Output: "In conclusion, renewable energy is essential for a sustainable future. By understanding its types and benefits, we can make informed decisions to support the transition to cleaner energy sources."

By chaining these prompts together, the model can create a complete essay in a structured and coherent manner.



Introduction to LangChain